# COMPUTING INFRASTRUCTURE FOR CLINICAL AI

Needs and challenges -- NSDF Summit on Federated Autonomous Science

Rudolph Pienaar -- Boston Children's Hospital / Harvard Medical School

rudolph@artemis: ~/nsdf2026

```
$> ./talk --budget=10m --venue=nsdf
```

# RUDOLPH PIENAAR

rudolph@artemis: ~ whoami

```
$> whoami --verbose
```

- Computational Scientist
- Systems Architect
- Brain Image Researcher

```
$> info --professional
```

- Assistant Professor of Radiology, Harvard Medical School
- Technical Director, Fetal-Neonatal Neuroimaging & Developmental Science Center (FNNDSC), Boston Children's
- PhD Biomedical Engineering (bio-control-model reinforcement learning) -- BEng / MEng

Created with slidedown

# WHAT I DO

rudolph@artemis: ~ whatido

```
$> whatido --at=bch
```

- Build the infrastructure that runs research computation **inside a hospital.**
- **ChRIS** -- open-source platform: containerised analyses, run next to clinical data.
- **HARBOR** -- a registry of **processable** data. Find the cohort, send the compute to it.
- Hospital on one side, research cloud on the other. **I live on the boundary.**

> THE HARD PART WAS NEVER THE MODEL. IT IS EVERYTHING AROUND IT.

# OUTLINE

rudolph@artemis: ~ outline

```
$> talk --outline

  [1] Current State of "AI" in Hospitals
  [2] What Exactly Is "AI"?
  [3] What Is Computing in a Hospital?
  [4] Governance
  [5] It's All About the Infrastructure
  [6] What Can Be Done?
  [7] The AI Bubble
  [8] Conclusions
```

# CURRENT STATE OF "AI" IN HOSPITALS

rudolph@artemis: ~ 01 state

```
$> ai_status --site=bch
```

- Honestly? Not much -- not in the Northeast, not at Boston Children's.
- I deployed the only semi-production AI workflows in clinical radiology at BCH.
- They are **predictive**: images in, labels and segmentations out.
- Generative AI is in use -- on **text**: notes, documentation, productivity.
- Where it touches diagnosis it is **advisory**: a human reads every output.

Generative AI at BCH: Becker's Hospital Review 2024 · OpenAI customer story, May 2026 (vendor-published)

# WHAT THE NUMBERS SAY

How much AI is in US healthcare in 2026 -- and doing what?

rudolph@artemis: ~ 01 state

```
$> survey --year=2026
    use any "AI" -- surveys don't say which kind; top uses are generative
    health systems      ▇▇▇
```

# CURRENT STATE OF "AI" IN HOSPITALS

What AI can do in a lab vs what hospitals actually run

rudolph@artemis: ~ 01 state

$> summary

- Deployed clinical AI: **predictive**, mostly imaging.
- Generative AI: notes and documentation, a human in the loop.
- Agents now beat specialists **in simulation** (88% vs 78%) -- yet finish only **46%** of real EHR tasks.

> "AI IN HOSPITALS" IS MUCH SMALLER THAN THE HEADLINES.

Ferber et al., Nature 2026 (MIRA, simulated patients) · Liu et al., PhysicianBench, arXiv 2605.02240 (2026, preprint)

# WHAT EXACTLY IS "AI"?

rudolph@artemis: ~ 02 what is ai

```
$> whatis ai --honest
```

- Today, "AI" means **LLMs**.
- But the AI delivering clinical value is **pattern matching**: tumour detection, tissue labelling, segmentation.
- **Predictive:** a deterministic forward pass. Rich input, bounded output.
- **Generative:** samples one plausible answer from many. The probabilism is the task.

# CAN LLMS READ IMAGES?

Frontier AI vs radiologists on 200 hard image cases (RadLE 2.0, Jul 2026)

rudolph@artemis: ~ 02 what is ai

```
$> benchmark --task=radiology --year=2026
```

score out of 2000: right answers earn points, confident wrong answers lose them




- 16 frontier models tested, incl. Gemini 3 Pro, GPT-5.6, Claude Fable 5. **None** reached the radiologists.
- Diffusion models **make and clean** images -- they don't read them.

> BEST MODEL: 23% BELOW RADIOLOGISTS -- AND OFTEN CONFIDENTLY WRONG.

CRASH Lab, RadLE 2.0 technical report, Jul 2026; scores via The Decoder, 19 Jul 2026 · Kazerouni et al., Med Image Anal 2023 (diffusion survey)

# WHAT EXACTLY IS "AI"?

rudolph@artemis: ~ 02 what is ai

```
$> diff predictive generative
```

- Same word, two machines -- they warrant **different trust**.
- This session's example of the first kind: small-cohort multiomics predictors (Alter).
- On "trustworthy autonomy": in the clinic the trust problem is not autonomy -- it is **telling the two apart.**

> CALL IT ALL "AI" AND YOU GOVERN IT ALL LIKE A CHATBOT.

# WHAT IS COMPUTING IN A HOSPITAL?

rudolph@artemis: ~ 03 compute

```
$> compute --inside=hospital
```

- For clinicians, until recently: Outlook and scheduling.
- Behind that: enterprise databases for billing.
- Computing as in **actual computing**? Barely existed. Still barely does.
- Forget AI -- most hospitals can't do **any** advanced computing.

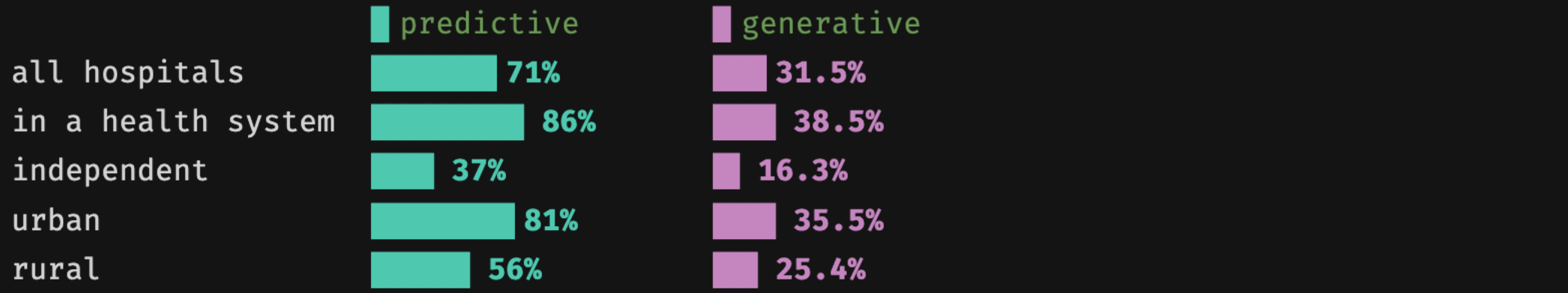
# RESOURCES, NOT REGIONS

Predictive vs generative AI, by hospital type

rudolph@artemis: ~ 03 compute

```
$> stats --ai --by=hospital --year=2024
```

% of US hospitals using each kind, built into the EHR (2024, same national survey)




predictive = ML or statistical risk models · generative = LLMs (ChatGPT, Gemini, DAX)

> THE HYPE IS GENERATIVE. THE USE IS PREDICTIVE. BOTH FOLLOW THE MONEY.

Predictive: ASTP Data Brief 80, Sep 2025 · Generative: Everson et al., JAMA Netw Open 2025, Table 1 · both from the 2024 AHA IT Supplement; imaging AI outside the EHR not counted

# WHAT IS COMPUTING IN A HOSPITAL?

Isn't the cloud the answer? Not yet.

rudolph@artemis: ~ 03 compute

```
$> cloud --status
> 80%+ use a public cloud. Few have left on-prem.
```

- "Move to the cloud" mostly means a hosted EHR and some workloads -- **not research computing.**
- 43% of healthcare IT leaders name the **skills gap** as the main cloud barrier.
- Planned 2026 spend on AI governance: large systems **6.8%** of IT and quality budget, small hospitals **2.3%**.

> THE GAP FOLLOWS MONEY AND SIZE.

KLAS via Becker's, Jul 2025 (n=36) · Rackspace State of Cloud 2025 (vendor, n=150) · Black Book via Becker's, Oct--Nov 2025 (n=182)

# GOVERNANCE

rudolph@artemis: ~ 04 governance

```
$> governance --apply --to=practice
```

- Well-meaning people who can't define AI, writing policy to govern it.
- Every policy assumes "AI" means generative AI.
- So it stalls on anything else.

# THE 100 MB MODEL

rudolph@artemis: ~ 04 governance

```
$> review tissue-seg.pt
     100 MB · images in · images out · runs on a tablet
  ▸  Q: hallucination guardrails? -- It does not sample. Deterministic forward pass.
  ▸  Q: PHI leaking to big model companies? -- There is no text. Nothing leaves.
  ▸  The process stalls anyway.
```

# GOVERNANCE

How ready are hospitals to govern AI? (health-system surveys, late 2025)

rudolph@artemis: ~ 04 governance

```
$> stats --governance
```

- Only **18%** of health systems have mature AI governance and a full strategy.
- AI submissions outpace committee decisions **~3:2** -- a growing backlog.
- Only **22%** could produce an AI audit trail within 30 days.

> SCRUTINY IS HEAVIEST ON THE SAFEST MODELS.

HFMA/Eliciting Insights, Aug 2025 (n=233) · Elion, The AI Governance Playbook, Nov 2025 (vendor) · Black Book via Becker's, Oct--Nov 2025 (n=182)

# IT'S ALL ABOUT THE INFRASTRUCTURE

rudolph@artemis: ~ 05 infrastructure

```
$> grep -r infrastructure ./clinical-ai
```

- The **first real need.**
- Places that didn't know computers could compute now need sophisticated computing.
- Cloud. Schedulers. Containers. Orchestration.
- Alien to IT departments built to install locked-down Windows desktops.

# THIS ROOM ALREADY KNOWS

rudolph@artemis: ~ 05 infrastructure

```
$> cat nsdf/themes.txt
```

- Federated data access. Provenance. Workflow portability. Governance.
- Medicine needs exactly these -- under stricter rules.
- One difference: **the data cannot move.**

# IT'S ALL ABOUT THE INFRASTRUCTURE

rudolph@artemis: ~ 05 infrastructure

```
$> echo $BOTTLENECK
> infrastructure
```

- The value is in small predictive models. Hospitals can't **run** them, and can't **approve** them.
- Infrastructure is the common fix.

> THE HARD PART WAS NEVER THE MODEL.

# WHAT CAN BE DONE?

rudolph@artemis: ~ 06 what can be done

```
$> plan --next
```

- **Triage governance by what a model is:** images or text? local or remote? deterministic or sampled?
- A fast lane for small, local, image-in/image-out models.
- **Shared infrastructure**, not a data centre per hospital: validated containers run next to the data (ChRIS).

# MOVE THE COMPUTATION

rudolph@artemis: ~ 06 what can be done

```
$> harbor --why
```

- Do not move the data. **Move the computation.**
- Find the cohort, send the compute to it. Only results come back -- no images, no PHI.
- A small hospital joins with a key, not a data centre.
- The precondition for federation: federated learning (Xu) needs sites that **can compute.**

> DO NOT MOVE THE DATA. MOVE THE QUESTION.

# WHAT CAN BE DONE?

rudolph@artemis: ~ 06 what can be done

```
$> ask --audience=nsdf
```

- Treat hospitals as **first-class sites** in national data fabrics.
- Networks to hospitals (Hart). Nodes that host clinical compute (Kirby). Shared standards (Musen).
- The hospital side brings the data and the questions.

> THE HOSPITALS LEAST ABLE TO COMPUTE NEED THIS MOST.

# THE AI BUBBLE

What tech conferences rarely say: the numbers don't add up

rudolph@artemis: ~ 07 bubble

$> cat ai_economics.txt

- Gartner, 14 Oct: AI data centres need $8.5T of revenue by 2030 to pay back; forecast is $3.6T. Over 80% of neoclouds gone by 2030.
- IMF, 13 Oct: 14 "circular" AI deals -- ~$950B of equity and debt in exchange for $1.4T of promised future revenue.
- Oracle: credit insurance at a record high (15 Oct); rated one notch above junk.
- OpenAI projects ~$278B of cash burn 2026--2030. Anthropic: cash-flow breakeven not before ~2028 (reported).
- Ed Zitron has made this case for two years. This week Gartner caught up.

Gartner via The Register, 14 Oct 2026 · IMF GFSR via TNW, 13 Oct 2026 · Oracle CDS: S&P/LSEG data via Securitas, 15 Oct 2026; S&P rating Jul 2026 · OpenAI: FT via Reuters, Sep 2026 · Anthropic: secondary reporting · Ed Zitron, Where's Your Ed At

# WHAT THE MONEY IS BETTING ON

The trillion-dollar valuations rest on three claims

rudolph@artemis: ~ 07 bubble

```
$> cat valuation_premise.txt
```

- **Claim:** AGI is close -- "we know how to build AGI" (Altman); "a country of geniuses in a datacenter" (Amodei).
- **Evidence:** frontier models trail radiologists on hard images and finish 46% of real EHR tasks.
- **Claim:** the machines are, or will become, conscious. **Evidence:** none -- consciousness science leans cautiously towards no for current AI.
- **Claim:** they solve what humans can't. Sep 2026: OpenAI says its AI proved Navier-Stokes blow-up -- a $1M Millennium Prize problem.
- **Look deeper:** two mathematicians had worked the same gap for a year, LLMs as tools. Formal proof and paper disagree. Clay still lists it **active**.

> A GOOD NEW TOOL, PRICED AS IF IT WERE A GOD.

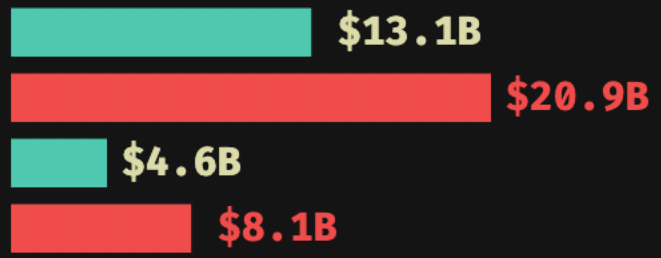
Altman, "Reflections", Jan 2025 · Amodei, "Machines of Loving Grace", Oct 2024 · CRASH Lab RadLE 2.0 · Liu et al., PhysicianBench 2026 · arXiv 2605.06965 (2026) · Navier-Stokes: OpenAI, Quanta, Axios, 8 Sep 2026; arXiv discrepancy paper, 6 Oct 2026; Clay Institute status Oct 2026

# EARNED VS LOST

OpenAI and Anthropic: revenue vs operating loss, 2025

rudolph@artemis: ~ 07 bubble

```
$> stats --labs --year=2025
    2025, $ billions -- what each company took in vs what it lost running the business
```




▸ Headline net losses were bigger ($38.5B, $42B) -- mostly one-off accounting charges, left out here.

**BOTH LOSE MORE THAN THEY EARN.**

Leaked draft IPO filings: OpenAI via Reuters/Yahoo Finance; Anthropic via Reuters/Fortune, 29 Sep 2026 · not yet public filings

# THE SUBSIDY

What the flat monthly plans hide: per-token vs flat, my own bills

rudolph@artemis: ~ 07 bubble

```
$> billing --compare
```

my own bills, similar coding workload




- Per month that's roughly **$15,000 vs $100** -- about 150x.
- Altman, Jan 2025: losing money on the $200 ChatGPT Pro plan -- "people use it much more than we expected."
- Anthropic, Jul 2025: weekly caps on Claude Code for users running it "continuously in the background, 24/7."

> TODAY'S FLAT PRICES ARE A SUBSIDY. BUDGETS BUILT ON THEM WON'T LAST.

Speaker's own billing: BCH enterprise Codex (per token) vs personal Claude Max (flat) -- different vendors, illustrative · Altman on X, 6 Jan 2025 (via Fortune) · Anthropic via TechCrunch, 28 Jul 2025

# THE GAP, IN TWO CHARTS

Revenue needed vs forecast -- and data centres delivered vs scheduled

rudolph@artemis: ~ 07 bubble

```
$> stats --ai-economics --to=2030
```

cumulative AI data-centre revenue to 2030, $ trillions







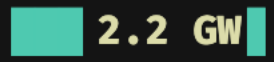
US data-centre capacity added per year, gigawatts










> THE REVENUE ISN'T THERE. THE BUILD-OUT RUNS BEHIND ITS OWN PLAN.

Gartner, 14 Oct 2026 (via The Register) · Goldman Sachs Research, 2026 (via secondary coverage): realized additions 2024-25, 2026 schedule, Q1 2026 actual; no full-year 2026 figure published yet

# IF IT POPS

What a crash would mean -- and why hospitals are, ironically, buffered

rudolph@artemis: ~ 07 bubble

```
$> what-if --crash
```

- Economy: a burst could cut UK GDP by **2.2%** (Bank of England); a correction would ripple through a "highly leveraged, circular" sector (IMF).
- Hospitals, ironically, are **buffered**: they barely use generative AI yet.
- Predictive AI just keeps running: 100 MB, on-prem, no frontier lab's balance sheet behind it.
- Who gets hurt: whoever just bought into generative AI -- repricing, failed vendors, stranded integrations.
- Science: a crash leaves cheap GPUs behind -- dark fibre after 2001, for federated compute.

> BUILD ON WHAT SURVIVES THE CRASH.

Bank of England Financial Stability Report, Jul 2026 (via Crypto Briefing) · IMF GFSR, Oct 2026 · adoption: Everson et al. 2025; ASTP 2025 · implications are the speaker's argument

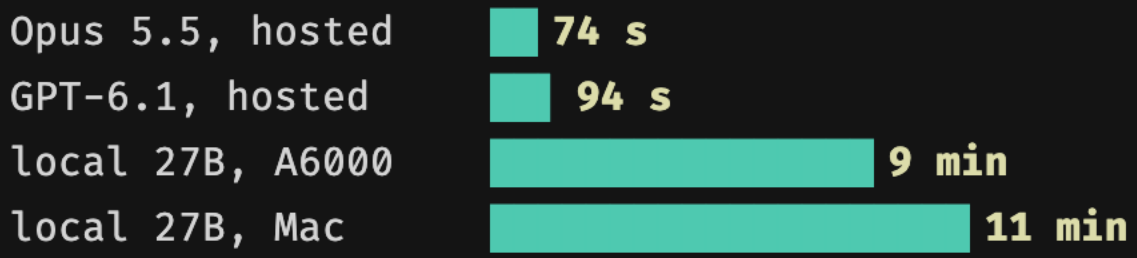
# THE AI BUBBLE

After the crash: models you can run yourself are catching up -- with a catch

rudolph@artemis: ~ 07 bubble

```
$> bench --task=7-bug-repair
```

my own study: median time to fully fix a 7-bug Python package (every setup below solved it)




- Local matched hosted on a bounded task -- **30 of 30** -- just 6-7x slower.
- The "4 months behind" figure is for open models on data-centre GPUs. On a workstation: smaller models, **smaller context windows**.
- Closing that gap takes enterprise GPUs -- **infrastructure** most hospitals don't have.

> GOOD ENOUGH FOR BOUNDED WORK. THE REST IS AN INFRASTRUCTURE PROBLEM.

Pienaar, "Local Language Models for Terminal Coding Agents in a Medium-Size Laboratory", unpublished (arXiv, in prep.), Oct 2026: 234 trials; qwen3.6:27b, 4-bit, 128K context · Epoch AI, 29 May 2026

# CONCLUSIONS

rudolph@artemis: ~ 08 conclusions

```
$> exit
> questions? I have opinions.
```

- The AI that works in hospitals is **small, predictive, and local.**
- Hospitals were built to run care, not computation.
- Governance written for LLMs blocks the models that already work.
- The fix: shared infrastructure, compute to the data -- and governance that can tell a segmentation model from a chatbot.